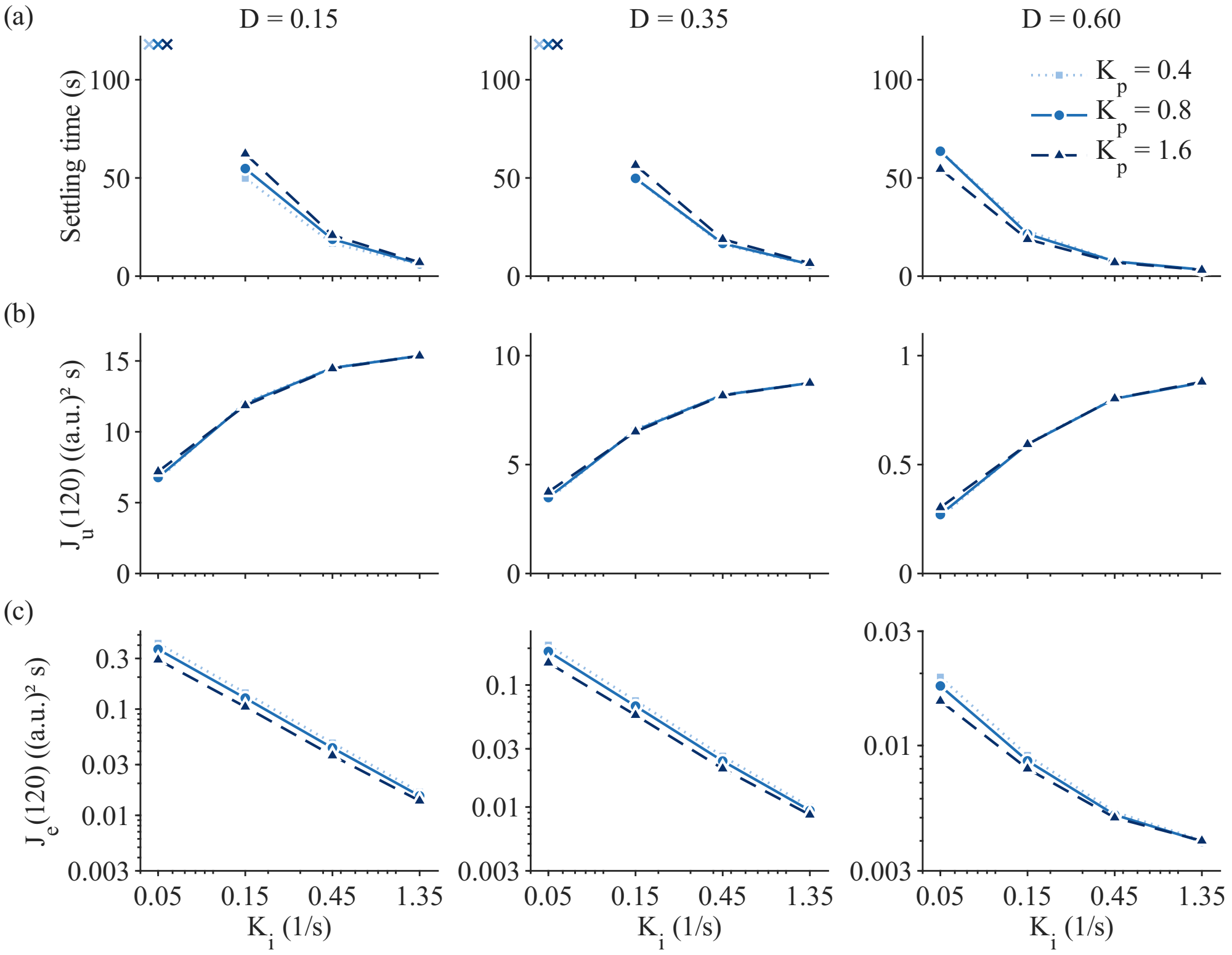
(a)
D = 0.15
D = 0.35
D = 0.60
Settling time (s)
100
50
0
K_p = 0.4
K_p = 0.8
K_p = 1.6
(b)
$J_u(120)$ ((a.u.)² s)
15
10
5
0
10
5
0
1
0.5
0
(c)
$J_e(120)$ ((a.u.)² s)
0.3
0.1
0.03
0.01
0.003
0.1
0.03
0.01
0.003
0.03
0.01
0.003
0.05
0.15
0.45
1.35
K_i (1/s)